

8	$F = N\rho AV = (27)(900)(0.4 \times 10^{-3})(3.0) = 0.087 \text{ N}$
9	$p_i = p_f$ $(1.0 \times 10^{27})(1.0 \times 10^4) - (1.0 \times 10^{30})u_s = (1.0 \times 10^{27})(-1.0 \times 10^4) + (-1.0 \times 10^{30})v_s$ $v_s - u_s = 20$